

$x(t)$ - input

$w(t)$ - kernel

Express how $x(t)$ changes with $w(t)$

Convolution * on Continuous Functions

$$(x * w)(t) = \int x(a) w(t-a) da$$

If $w(t)$ is a window function

$(x * w)(t)$ is weighted moving average of $x(t)$

Convolution * on Discrete Functions

$$(x * w)(t) = \sum_{a=-\infty}^{\infty} x(a) w(t-a)$$

Alternatively,

$$(x * w)(t) = \sum_{a=-\infty}^{\infty} x(t+a) w(a)$$

Choose $w(t)$ to be non-zero only near the origin.

$$\text{non-zero} \quad -c \leq t \leq c$$

the original...

$$w(t) = \begin{cases} \text{non-zero} & -c \leq t \leq c \\ 0 & \text{otherwise} \end{cases} \quad c - \text{constant}$$

Width - # of non-zero elements in kernel

Example

Compute $(x * w)(t)$ when

$$w(t) = \begin{cases} 1 & \text{if } -2 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} (x * w)(5) &= w(-2)x(5-2) + w(-1)x(5-1) \\ &\quad + w(0)x(5+0) + w(1)x(5+1) \\ &\quad + w(2)x(5+2) \\ &= x(3) + x(4) + x(5) + x(6) + x(7) \end{aligned}$$

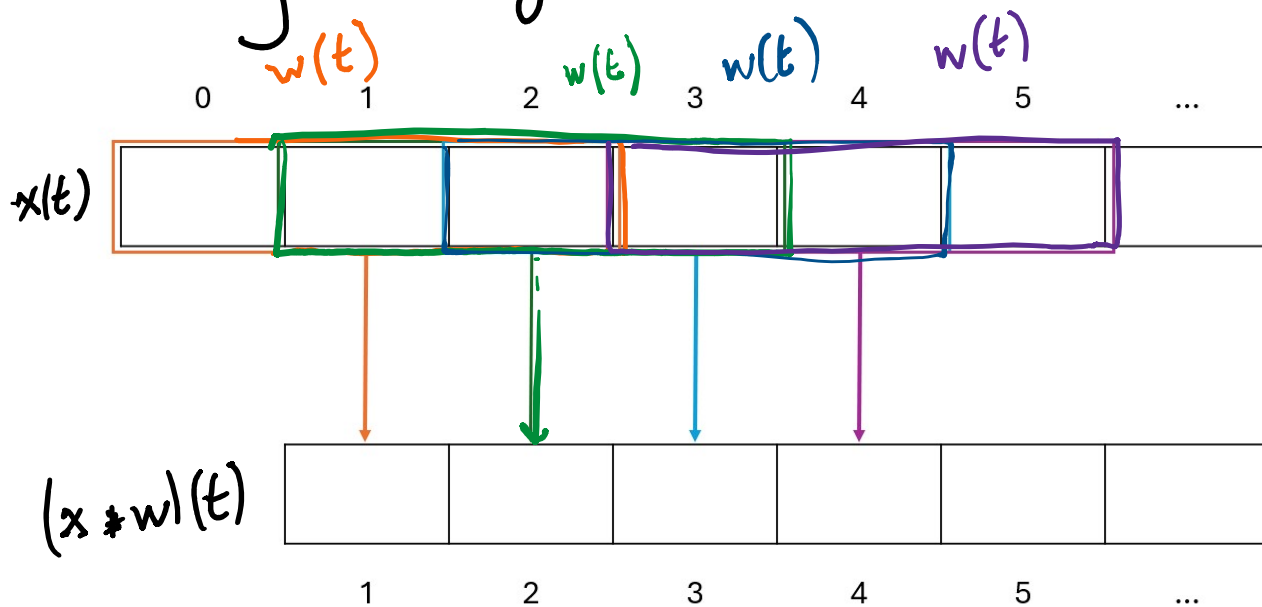
$$\begin{aligned} (x * w)(-2) &= x(-4) + x(-3) + x(-2) \\ &\quad + x(-1) + x(0) \end{aligned}$$

$$\begin{aligned} (x * w)(t) &= x(t-2) + x(t-1) + x(t) \\ &\quad + x(t+1) + x(t+2) \end{aligned}$$

Kernel's width is 5.

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Thinking visually ...



Kernel width 3

Example

$$w(t) = \begin{cases} 1 & \text{if } -1 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Kernel width 3

